INCH-POUND

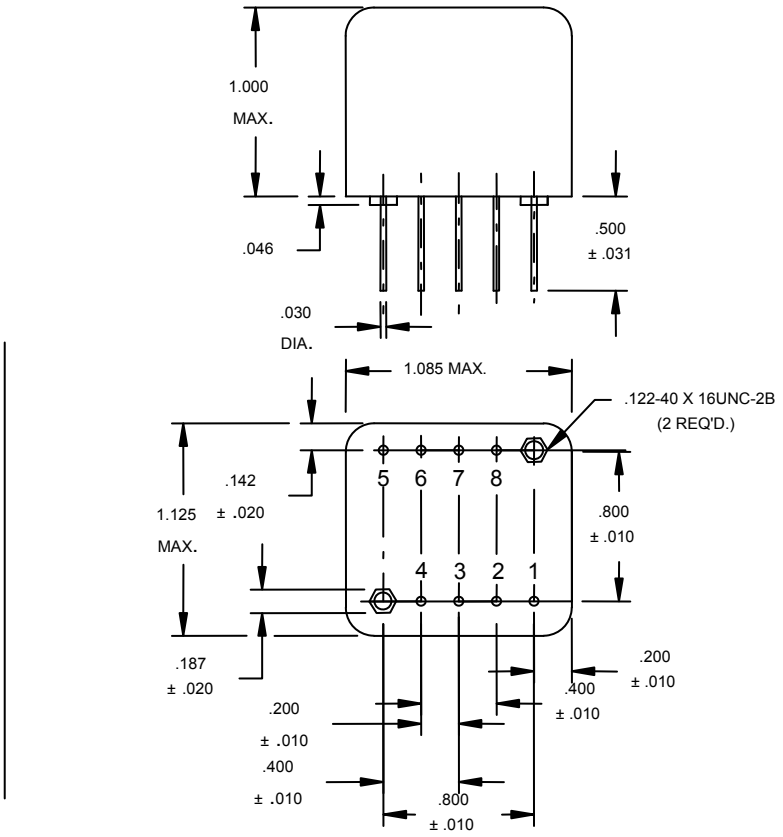
MIL-PRF-27/79A
9 November 2009
SUPERSEDING
MIL-PRF-27/79
29 August 1975

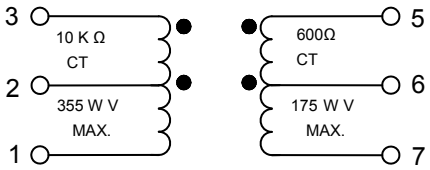
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-PRF-27.





CIRCUIT DIAGRAM AND MARKING

Inches	mm
.010	.254
.020	.508
.030	.762
.031	.7874
.045	1.143
.142	3.6068
.187	4.7498
.200	5.08
.400	10.16
.500	12.70
.800	20.32
1.000	25.40
1.085	27.56
1.125	28.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on top of the case.

FIGURE 1. Dimensions and configuration

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REQUIREMENTS: (When numbers in parentheses, such as (1.-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Source impedance (1-3): 10k ohms ct ±10 percent.

Load impedance (5-7): 600 ohms ct.

Primary power level: 6 watts maximum (200 Hz to 10 kHz)
25 watt (60 Hz to 200 Hz)

Working voltage:

(1-3) 355 volts maximum.
(5-7) 175 volts maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy glass.

Terminals: Pin type, Dumet, hot tin dipped.

Weight: 2.5 ounces, maximum.

Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage (at sea level): 1,000 volts rms (1-3) and 500 volts rms (5-7)

Electrical characteristics:

Harmonic distortion: Not more than ±10 percent at 6 watts, 200 Hz to 10 kHz across (5-7).

Primary impedance (1-3): 10k ohms ct ±10 percent at 1 kHz.

Secondary load impedance (5-7): 600 ohms ct.

Frequency response: 60 Hz to 10 kHz ±1 dB maximum.

DC resistance:

Primary (1-3): 700 ohms ±25 percent.
Secondary (5-7): 50 ohms ±25 percent.

No load (center tap (ct) voltage unbalance only):

With 50 volts, 60 Hz and 0 amp voltage unbalance dc applied to (1-3)

Unbalance (1-2) and (2-3): ±1 percent maximum.
Unbalance (5-6) and (6-7): ±1 percent maximum.

Polarity: Additive, with terminals 3 and 5 connected.

Temperature rise: 45°C.

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Vibration, high frequency: MIL-STD-202, method 204, test condition B.

Shock (specified pulse): MIL-STD-202, method 213, test condition H.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/79-01

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5950-2009-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at http://assist.daps.dla.mil.

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